Cloud Cost Optimisation
A Primer

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6 October 2021

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01 Challenge

Let’s talk about…

The Cloud

Cloud Market

	revenue (USD)	market share (%)	YoY growth (%)
Amazon Web Services	26,20	40.8	28.7
Microsoft Azure	12,69	19.7	59.2
Alibaba Cloud	6,12	9.5	52.8
Google Cloud Platform	3,93	6.1	202.8
Others	15,34	24	25.6

Source: gtnr.it/3B9xxhE

Cloud Market

$64B

Annual spend on public cloud

Source: gtnr.it/3B9xxhE

Cloud Market

$64B

Annual spend on public cloud

41%

Annual growth of public cloud

Source: gtnr.it/3B9xxhE

Cloud Waste

$64B

Annual spend on public cloud

$17.6B

Annual public cloud waste

Source: devops.com/the-cloud-is-booming-but-so-is-cloud-waste

Cloud Waste

$64B

Annual spend on public cloud

$17.6B

Annual public cloud waste

20-30%

Spend attributed to cloud waste

Source: flexera.com/blog/cloud/cloud-computing-trends-2021-state-of-the-cloud-report

Cloud Waste

“Between idle and over-provisioned resources alone, that’s $17.6B in cloud spend that will be completely wasted this year. And the potential is even higher […]

That’s a minimum of about **$5M wasted *per day***, every day this year, that could be reallocated toward other areas of the business.”

$11B

Annual public cloud waste

$6.6B

Wasted from over-sized resources

Source: devops.com/the-cloud-is-booming-but-so-is-cloud-waste

Cloud Waste Side-Effects

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Reduced Productivity

Wasted spend is correlated with **poor standardisation**, or worse yet: anti-patterns.

If there’s no easy way for developers to re-use patterns, then everybody **re-invents the wheel**, over and over again.

Cloud Waste Side-Effects

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Risk

Lack of governance and oversight across the enterprise results in an **increased risk surface**.

Validated models are easier to reason about, which is part of a heightened security posture.

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Misallocation

Literally: money that could - and should - be spent on other things.

Brass tacks: wasted cloud spend is basically a vampire that is **making your organisation weaker**.

Let’s talk about…

Opportunity

Shift-left on Cloud Spend

Engineers as the new cloud financial controllers

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- Availability of billing minutiae

Shift-left on Cloud Spend

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- Wait a minute… an API full of **metrics**?

Shift-left on Cloud Spend

Engineers as the new cloud financial controllers

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- That can be **correlated** with deployments?

Shift-left on Cloud Spend

Engineers as the new cloud financial controllers

- Availability of billing minutiae
- Billing data is just that: **data**
- Programmatic access to that **data**
- Wait a minute… an API full of **metrics**?
- That can be **correlated** with deployments?
- This sounds like a job for **dashboards**!

Shift-left on Cloud Spend

…y tho?

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- Get **involved** in product discussions

Shift-left on Cloud Spend

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- Bring **insight** to business decisions

Shift-left on Cloud Spend

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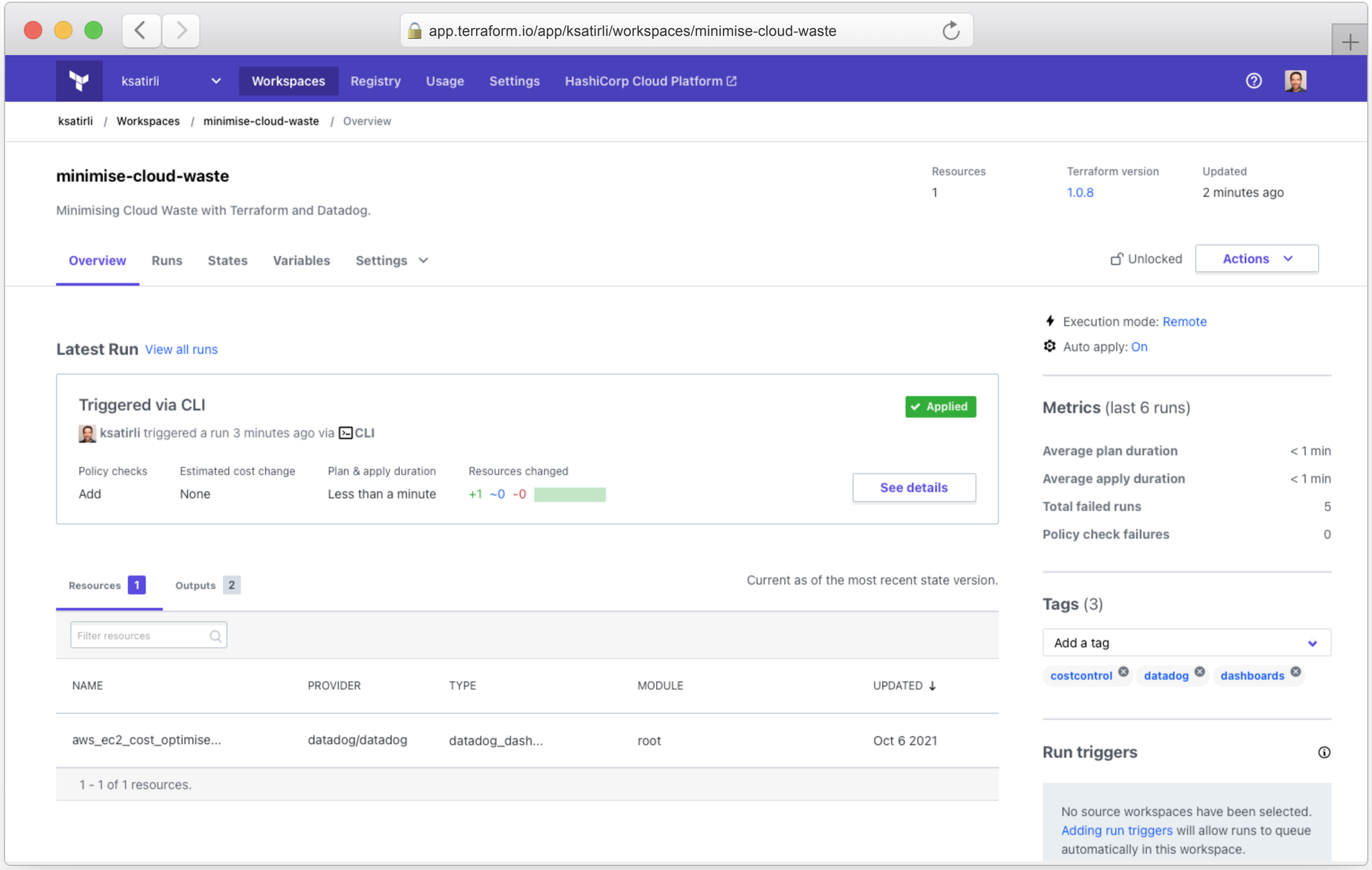
- Increase your capacity to generate **value**

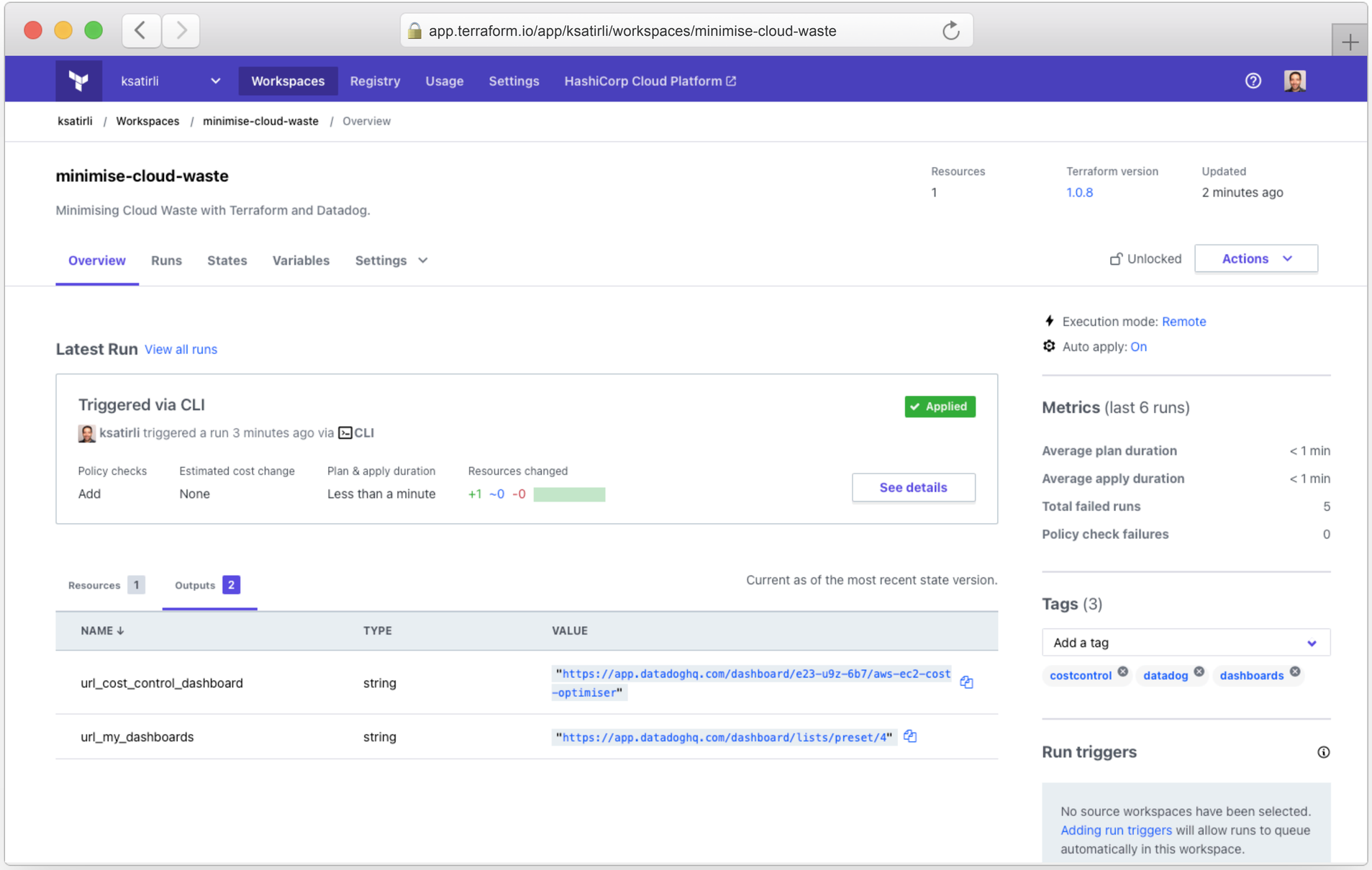
Shift-left on Cloud Spend

…y tho?

- Get **involved** in product discussions
- Bring **insight** to business decisions
- Increase your capacity to generate **value**
- Level-up your **career**

Demo


app.terraform.io/app/ksatirli/workspaces/minimise-cloud-waste
ksatirli
Workspaces
Registry
Usage
Settings
HashiCorp Cloud Platform
ksatirli / Workspaces / minimise-cloud-waste / Overview
minimise-cloud-waste
Minimising Cloud Waste with Terraform and Datadog.
Resources
1
Terraform version
1.0.8
Updated
2 minutes ago
Overview
Runs
States
Variables
Settings
Unlocked
Actions
Latest Run
View all runs
Triggered via CLI
Applied
ksatirli triggered a run 3 minutes ago via CLI
Policy checks
Add
Estimated cost change
None
Plan & apply duration
Less than a minute
Resources changed
+1 ~0 -0
See details
Execution mode: Remote
Auto apply: On
Metrics (last 6 runs)
Average plan duration
< 1 min
Average apply duration
< 1 min
Total failed runs
5
Policy check failures
0
Resources 1
Outputs 2
Current as of the most recent state version.
Filter resources
NAME
PROVIDER
TYPE
MODULE
UPDATED
aws_ec2_cost_optimise...
datadog/datadog
datadog_dash...
root
Oct 6 2021
1 - 1 of 1 resources.
Tags (3)
Add a tag
costcontrol
datadog
dashboards
Run triggers
No source workspaces have been selected.
Adding run triggers will allow runs to queue automatically in this workspace.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< 1 min
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5
Policy check failures
0
Resources 1
Outputs 2
Current as of the most recent state version.
NAME ↓
TYPE
VALUE
url_cost_control_dashboard
string
"https://app.datadoghq.com/dashboard/e23-u9z-6b7/aws-ec2-cost-optimiser"
url_my_dashboards
string
"https://app.datadoghq.com/dashboard/lists/preset/4"
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datadog
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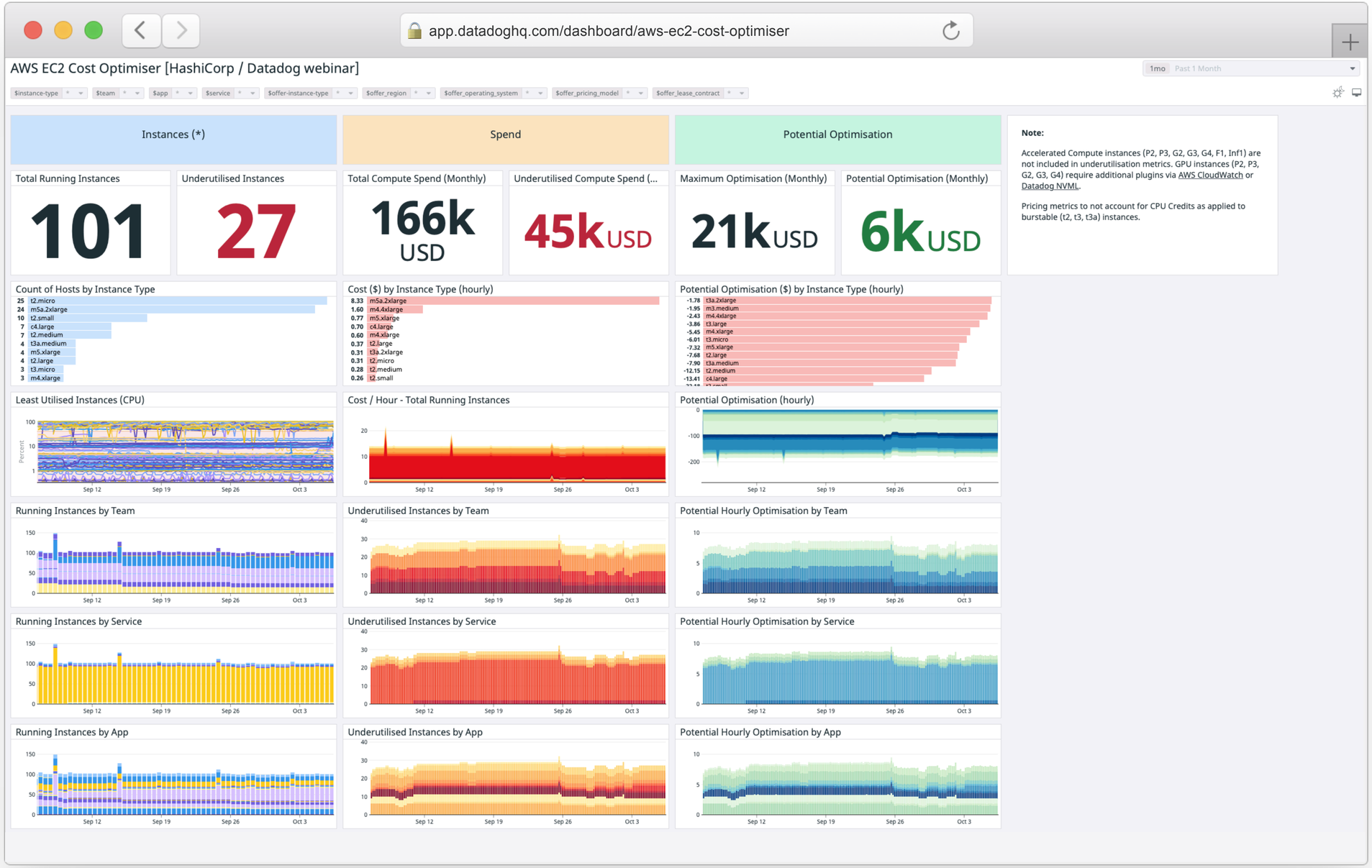
app.datadoghq.com/dashboard/aws-ec2-cost-optimiser
AWS EC2 Cost Optimiser [HashiCorp / Datadog webinar]
1mo Past 1 Month
$instance-type $team $app $service $offer-instance-type $offer_region $offer_operating_system $offer_pricing_model $offer_lease_contract
Instances (*)
Spend
Potential Optimisation
Total Running Instances
101
Underutilised Instances
27
Total Compute Spend (Monthly)
166k USD
Underutilised Compute Spend (...
45k USD
Maximum Optimisation (Monthly)
21k USD
Potential Optimisation (Monthly)
6k USD
Note:
Accelerated Compute instances (P2, P3, G2, G3, G4, F1, Inf1) are not included in underutilisation metrics. GPU instances (P2, P3, G2, G3, G4) require additional plugins via AWS CloudWatch or Datadog NVML.
Pricing metrics to not account for CPU Credits as applied to burstable (t2, t3, t3a) instances.
Count of Hosts by Instance Type
25 t2.micro
24 m5a.2xlarge
10 t2.small
7 c4.large
7 t2.medium
4 t3a.medium
4 m5.xlarge
4 t2.large
3 t3.micro
3 m4.xlarge
Cost ($) by Instance Type (hourly)
8.33 m5a.2xlarge
1.60 m4.4xlarge
0.77 m5.xlarge
0.70 c4.large
0.60 m4.xlarge
0.37 t2.large
0.31 t3a.2xlarge
0.31 t2.micro
0.28 t2.medium
0.26 t2.small
Potential Optimisation ($) by Instance Type (hourly)
-1.78 t3a.2xlarge
-1.95 m3.medium
-2.43 m4.4xlarge
-3.86 t3.large
-5.45 m4.xlarge
-6.01 t3.micro
-7.32 m5.xlarge
-7.68 t2.large
-7.90 t3a.medium
-12.15 t2.medium
-13.41 c4.large
Least Utilised Instances (CPU)
Percent
Sep 12 Sep 19 Sep 26 Oct 3
Cost / Hour - Total Running Instances
Potential Optimisation (hourly)
Running Instances by Team
Underutilised Instances by Team
Potential Hourly Optimisation by Team
Running Instances by Service
Underutilised Instances by Service
Potential Hourly Optimisation by Service
Running Instances by App
Underutilised Instances by App
Potential Hourly Optimisation by App

Thank You!

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